



Multiobjective Optimization

Optimization of Complex Systems – March 9th 2026

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Introduction

$$\min f_1(x), \dots, f_p(x)$$

$$\text{s.t. } x \in \mathcal{F}.$$

$$x \in \mathbb{R}^n, p > 1, f_i : \mathbb{R}^n \rightarrow \mathbb{R}, \mathcal{F} \subseteq \mathbb{R}^n.$$

- $\mathbb{R}^n$ is the **decision space**
- $\mathbb{R}^p$ is the **objective space**
- $x \in \mathbb{R}^n$ is a **decision vector**
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Introduction

We define:

- $f(x) = (f_1(x), \dots, f_p(x)) : \mathbb{R}^n \rightarrow \mathbb{R}^p$ **objective vector function**
- $\mathcal{Z} = f(\mathcal{F})$ **feasible objective region:**

$$\mathcal{Z} = \{z \in \mathbb{R}^p : z = f(x), x \in \mathcal{F}\}.$$

- z^{id} **ideal objective vector:**

$$z_i^{id} = \min_{x \in \mathcal{F}} f_i(x), \quad i = 1, \dots, p.$$

Note: we assume $z^{id} \notin \mathcal{Z}$, i.e. the objective functions are *in conflict* with each other.

Ordering in p -dimensional space

We can define on $\mathbb{R}^p$ a **partial strict ordering**, known as **Pareto dominance**.

Named after the Italian economist and sociologist **Vilfredo Pareto** (1848–1923), who first introduced the concept of efficient allocation in economics.



Ordering in p -dimensional space

Given two vectors z^1 and z^2 in $\mathbb{R}^p$, we say that:

z^1 *dominates* (in the Pareto sense) z^2 , written $z^1 \preceq z^2$, if

$z_i^1 \leq z_i^2$ for **every** $i = 1, \dots, p$, and

$z_j^1 < z_j^2$ for **some** $j \in \{1, \dots, p\}$.

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$$z_j^1 < z_j^2 \text{ for some } j \in \{1, \dots, p\}.$$

Note: the ordering is **only partial** — there exist pairs z^1, z^2 such that **neither**

$$z^1 \preceq z^2 \quad \text{nor} \quad z^2 \preceq z^1.$$

In this case, z^1 and z^2 are said to be **mutually non-dominated**.

Weak dominance

Given two vectors z^1 and z^2 in $\mathbb{R}^p$, we say that:

z^1 *weakly dominates* (in the Pareto sense) z^2 , written $z^1 \prec z^2$, if

$$z_i^1 < z_i^2 \text{ for every } i = 1, \dots, p.$$

Note: if $z^1 \prec z^2$ then $z^1 \preceq z^2$, but the converse does **not** hold in general.

Pareto optimality

Given the multiobjective problem

$$\min f_1(x), \dots, f_p(x) \quad \text{s.t. } x \in \mathcal{F},$$

a point $x^* \in \mathcal{F}$ is **Pareto optimal** if there exists no other $x \in \mathcal{F}$ such that

$$f(x) \preceq f(x^*).$$

In the objective space, the set of Pareto optimal points Ω_P is known as the:

- efficient frontier, or
- Pareto front.

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Weak Pareto optimality

A point $x^* \in \mathcal{F}$ is **weakly Pareto optimal** if there exists no other $x \in \mathcal{F}$ such that

$$f(x) < f(x^*).$$

Note: the set of Pareto optimal points (Ω_P) is contained in the set of weakly Pareto optimal points (Ω_D):

$$\Omega_P \subseteq \Omega_D.$$

Example (1) — finding non-dominated points

Determine the set of **non-dominated** points among the following:

	a	b	c	d	e	f	g	h	i	j
$f_1(x)$	4	9	4	3	2	3	5	8	3	5
$f_2(x)$	2	10	5	10	6	3	9	1	9	9

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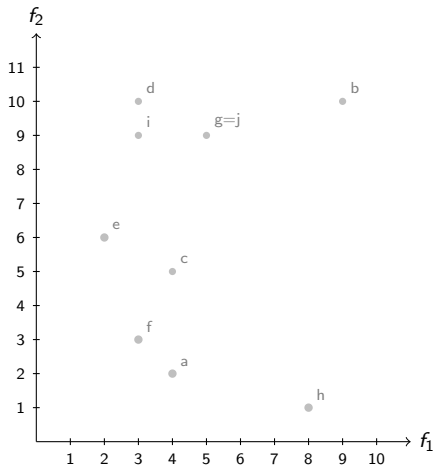
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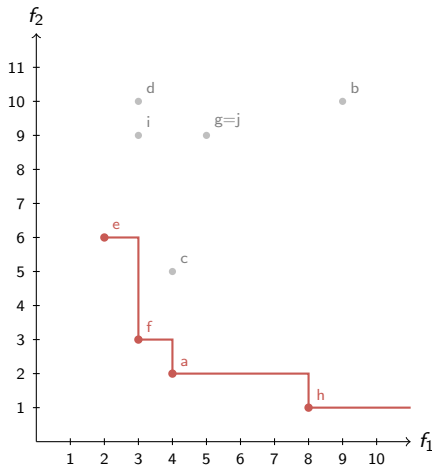
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Example (1) — Pareto front



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Non-dominated points:

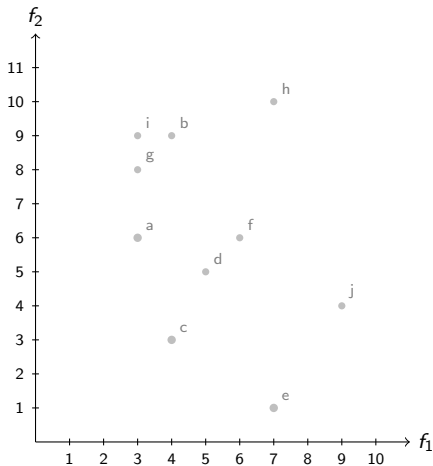
Point	(f_1, f_2)
a	$(4, 2)$
e	$(2, 6)$
f	$(3, 3)$
h	$(8, 1)$

Example (2) — finding non-dominated points

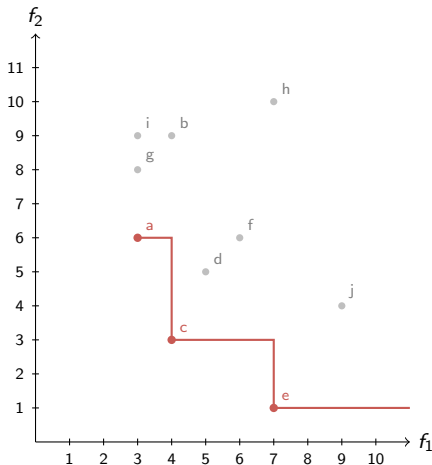
Determine the set of **non-dominated** points among the following:

	a	b	c	d	e	f	g	h	i	j
f_1	3	4	4	5	7	6	3	7	3	9
f_2	6	9	3	5	1	6	8	10	9	4

Example (2) — Pareto front



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Non-dominated points:

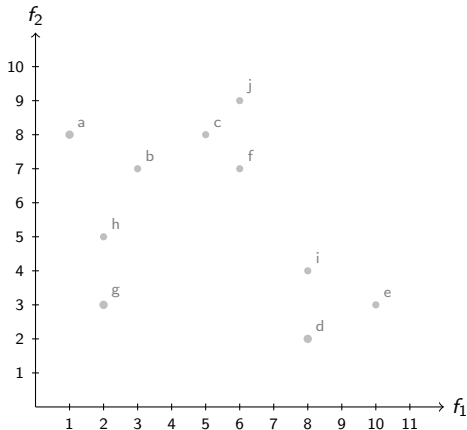
	a	c	e
f_1	3	4	7
f_2	6	3	1

Example (3) — finding non-dominated points

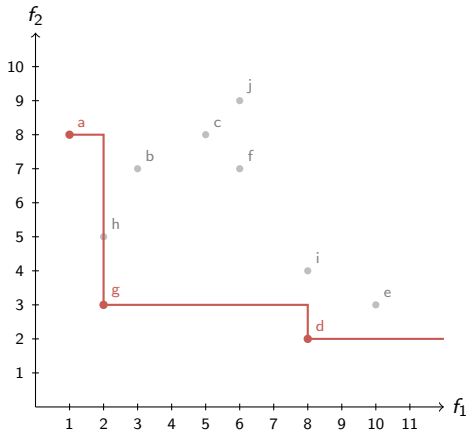
Determine the set of **non-dominated** points among the following:

	a	b	c	d	e	f	g	h	i	j
f_1	1	3	5	8	10	6	2	2	8	6
f_2	8	7	8	2	3	7	3	5	4	9

Example (3) — Pareto front



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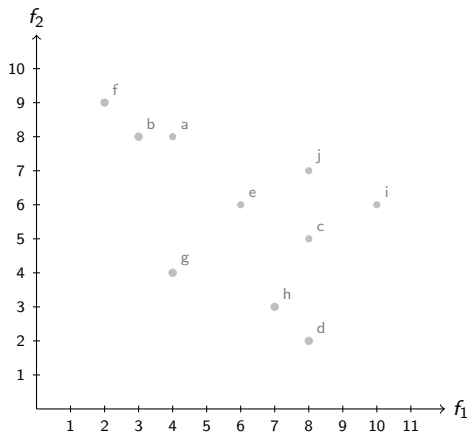
	a	d	g
f_1	1	8	2
f_2	8	2	3

Example (4) — finding non-dominated points

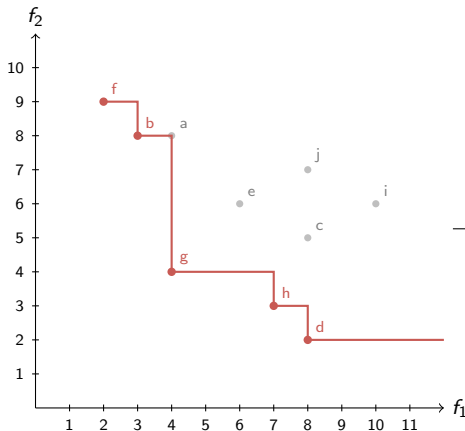
Determine the set of **non-dominated** points among the following:

	a	b	c	d	e	f	g	h	i	j
f_1	4	3	8	8	6	2	4	7	10	8
f_2	8	8	5	2	6	9	4	3	6	7

Example (4) — Pareto front



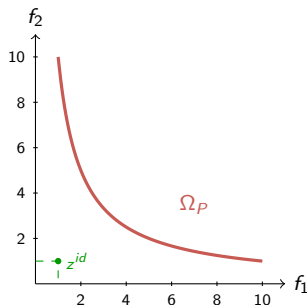
Example (4) — Pareto front



Non-dominated points:

	b	d	f	g	h
f_1	3	8	2	4	7
f_2	8	2	9	4	3

Example (5) — continuous Pareto front

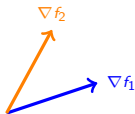


Key observation: in difficult problems the objectives cannot be simultaneously minimized.

Descent directions in the multiobjective case

In single-objective optimization, d is a descent direction at x if $\nabla f(x)^\top d < 0$.

What changes with multiple objectives? We need $\nabla f_i(x)^\top d < 0$ for *all* $i = 1, \dots, p$ simultaneously. This may or may not be possible depending on the geometry of the gradients.

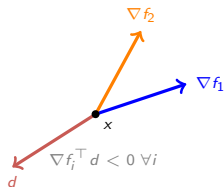


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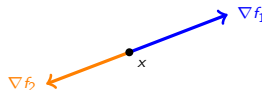
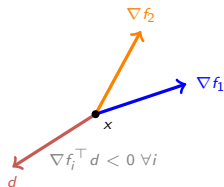


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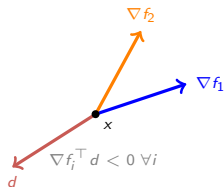


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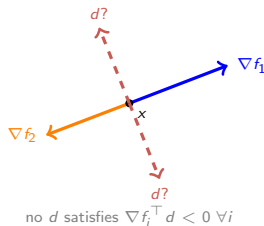
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Case 1: descent direction exists



Case 2: no descent direction



Common descent directions

Definition (Common descent direction)

Given $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ continuously differentiable, a direction $d \in \mathbb{R}^n$, $d \neq 0$, is a **common descent direction** for f at x if

$$\nabla f_i(x)^\top d < 0, \quad i = 1, \dots, p.$$

Geometric interpretation. d must lie in the open half-space $\{d : \nabla f_i(x)^\top d < 0\}$ for every i simultaneously, i.e., in the intersection:

$$\mathcal{D}(x) = \bigcap_{i=1}^p \{d \in \mathbb{R}^n : \nabla f_i(x)^\top d < 0\}.$$

Question: When is $\mathcal{D}(x)$ non-empty? What geometric condition on the gradients $\nabla f_1(x), \dots, \nabla f_p(x)$ guarantees this?

$\mathcal{D}(x) \neq \emptyset$ if and only if 0 does not belong to the convex hull of the gradients, i.e.,

$$0 \notin \text{conv}\{\nabla f_1(x), \dots, \nabla f_p(x)\}.$$

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Unconstrained multiobjective problem

We now focus on the **unconstrained** case:

$$\min_{x \in \mathbb{R}^n} f(x) = (f_1(x), \dots, f_p(x))^{\top}$$

with $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ continuously differentiable for all $i = 1, \dots, p$.

Definition (Descent direction set)

Given $x \in \mathbb{R}^n$, define:

$$\mathcal{D}_P(x) = \{d \in \mathbb{R}^n : d \neq 0, f(x + \beta d) \preceq f(x) \forall \beta \in (0, \alpha], \alpha > 0\}.$$

$\mathcal{D}_P(x)$ is the set of directions along which f **does not increase** (in the Pareto sense) for small steps.

The set $\mathcal{D}(x)$ defined earlier satisfies:

$$\mathcal{D}(x) = \{d \in \mathbb{R}^n : \nabla f_i(x)^{\top} d < 0, i = 1, \dots, p\} \subseteq \mathcal{D}_P(x).$$

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Necessary condition for Pareto optimality

Proposition (Necessary condition)

If $x^ \in \mathbb{R}^n$ is a (local) Pareto optimum of the unconstrained problem, then*

$$\mathcal{D}_P(x^*) = \emptyset,$$

i.e., no common descent direction exists at x^ .*

Since $\mathcal{D}(x) \subseteq \mathcal{D}_P(x)$, a necessary condition is also:

$$\mathcal{D}(x^*) = \emptyset, \quad \text{i.e.,} \quad 0 \in \text{conv}\{\nabla f_1(x^*), \dots, \nabla f_p(x^*)\}.$$

This means: at a Pareto optimum, the gradients must **surround the origin** — it is impossible to find a direction that simultaneously decreases all objectives.

Question: How does this reduce to the familiar condition $\nabla f(x^*) = 0$ when $p = 1$?

Necessary condition for Pareto optimality

Proposition (Necessary condition)

If $x^ \in \mathbb{R}^n$ is a (local) Pareto optimum of the unconstrained problem, then*

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Since $\mathcal{D}(x) \subseteq \mathcal{D}_P(x)$, a necessary condition is also:

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The stationarity measure $\theta(x)$

We quantify how far x is from being Pareto optimal via:

Definition (Stationarity measure)

$$\theta(x) = \min_{\|d\|=1} \max_{i=1,\dots,p} \nabla f_i(x)^\top d.$$

Key properties:

- $\theta(x) \leq 0$ for all x (can always find d decreasing at least one f_i)
- $\theta(x) = 0 \iff \mathcal{D}(x) = \emptyset \iff x$ is a **Pareto stationary point**
- $\theta(x) < 0 \implies x$ is **not** a local Pareto optimum: the direction

$$d(x) = \arg \min_{\|d\|=1} \max_i \nabla f_i(x)^\top d$$

is a common descent direction

Connection to single-objective case ($p = 1$):

$$\theta(x) = \min_{\|d\|=1} \nabla f(x)^\top d = -\|\nabla f(x)\|.$$

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Necessary optimality condition (Pareto)

Theorem (Necessary condition for Pareto optimality)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ be continuously differentiable. If x^* is a local Pareto optimum, then x^* is a **Pareto stationary point**:

$$\theta(x^*) = 0,$$

equivalently, $0 \in \text{conv}\{\nabla f_1(x^*), \dots, \nabla f_p(x^*)\}$, i.e., there exist $\lambda_1, \dots, \lambda_p \geq 0$ with $\sum_{i=1}^p \lambda_i = 1$ such that

$$\sum_{i=1}^p \lambda_i \nabla f_i(x^*) = 0.$$

Interpretation: the gradients are **positively balanced** at the optimum — any unit-norm movement from x^* must increase at least one objective.

Remark: when $p = 1$, this reduces to $\lambda_1 \nabla f(x^*) = 0$ with $\lambda_1 = 1$, i.e., $\nabla f(x^*) = 0$ — the standard first-order condition.

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Necessary condition for weak Pareto optimality

Definition (Weak descent direction set)

$$\mathcal{D}_w(x) = \{d \in \mathbb{R}^n : d \neq 0, f(x + \beta d) \prec f(x) \forall \beta \in (0, \alpha], \alpha > 0\}.$$

Since $\mathcal{D}(x) \subseteq \mathcal{D}_w(x)$, the same argument applies:

Proposition

If x^ is a (local) weakly Pareto optimal point, then $\mathcal{D}(x^*) = \emptyset$, i.e.,*

$$\theta(x^*) = 0.$$

Remark: Pareto optimality $\Rightarrow$ weak Pareto optimality $\Rightarrow$ Pareto stationarity. The converse implications do **not** hold in general.

$$\Omega_P \subseteq \Omega_D \subseteq \{x : \theta(x) = 0\}.$$

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Summary: descent direction sets

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ be continuously differentiable and let $x \in \mathbb{R}^n$.

Pareto: $\mathcal{D}_P(x) = \{d \in \mathbb{R}^n : d \neq 0, f(x + \beta d) \preceq f(x) \forall \beta \in (0, \alpha], \exists \alpha > 0\}$

Weak: $\mathcal{D}_w(x) = \{d \in \mathbb{R}^n : d \neq 0, f(x + \beta d) \prec f(x) \forall \beta \in (0, \alpha], \exists \alpha > 0\}$

First-order: $\mathcal{D}(x) = \{d \in \mathbb{R}^n : \nabla f_i(x)^\top d < 0, i = 1, \dots, k\}$

Inclusion structure:

$$\mathcal{D}(x) \subseteq \mathcal{D}_P(x) \subseteq \mathcal{D}_w(x)$$

Optimality via empty cones:

Optimality type	Necessary condition	Equivalent to
Pareto optimum	$\mathcal{D}_P(x^*) = \emptyset$	$\Rightarrow \mathcal{D}(x^*) = \emptyset$
Weak Pareto optimum	$\mathcal{D}_w(x^*) = \emptyset$	$\Rightarrow \mathcal{D}(x^*) = \emptyset$
Pareto stationary	$\mathcal{D}(x^*) = \emptyset$	$\theta(x^*) = 0$